

Chapter 9: Introduction to Calculus

Limits, Derivatives, Differentiation Rules & Definite Integration

9.1 Limits & Derivative Definition

Calculus studies continuous change. The derivative represents the instantaneous rate of change or tangent slope.

Definition of Derivative: $f'(x) = \lim_{h \rightarrow 0} [f(x+h) - f(x)] / h$

Basic Differentiation Rules

- *Power Rule:* $d/dx(x^n) = n \cdot x^{n-1}$
- *Exponential:* $d/dx(e^x) = e^x$
- *Natural Log:* $d/dx(\ln x) = 1/x$
- *Sine & Cosine:* $d/dx(\sin x) = \cos x$, $d/dx(\cos x) = -\sin x$

Product & Quotient Rules

- *Product Rule:* $(u \cdot v)' = u'v + uv'$
- *Quotient Rule:* $(u/v)' = (u'v - uv') / v^2$
- *Chain Rule:* $d/dx[f(g(x))] = f'(g(x)) \cdot g'(x)$

Worked Example 9.1: Chain Rule Differentiation

Problem: Find the derivative of $f(x) = (3x^2 - 5)^4$.

Solution: Apply Chain Rule with inner function $u = 3x^2 - 5$:

$$f'(x) = 4(3x^2 - 5)^3 \cdot d/dx(3x^2 - 5)$$

$$f'(x) = 4(3x^2 - 5)^3 \cdot (6x) = \mathbf{24x(3x^2 - 5)^3}.$$

9.2 Integration & Fundamental Theorem of Calculus

Integration is the inverse operation of differentiation, representing the area under a curve.

Indefinite Power Rule: $\int x^n dx = (x^{n+1} / (n + 1)) + C \quad (n \neq -1)$

Fundamental Theorem of Calculus: $\int_a^b f(x) dx = F(b) - F(a) \quad \text{where } F'(x) = f(x)$

Worked Example 9.2: Definite Integral Calculation

Problem: Evaluate the definite integral $\int_1^3 (3x^2 + 2x) dx$.

Solution: Find the antiderivative $F(x)$:

$$F(x) = 3(x^3 / 3) + 2(x^2 / 2) = x^3 + x^2.$$

Evaluate from 1 to 3:

$$F(3) = (3)^3 + (3)^2 = 27 + 9 = 36.$$

$$F(1) = (1)^3 + (1)^2 = 1 + 1 = 2.$$

$$\text{Area} = F(3) - F(1) = 36 - 2 = \mathbf{34}.$$

Chapter 9 Review & Practice Problems

1. Evaluate the limit: $\lim_{x \rightarrow 2} (x^2 - 4) / (x - 2)$.
2. Differentiate $y = x^3 \cdot e^{2x}$ using the Product and Chain rules.
3. Find the slope of the tangent line to $f(x) = \sin(x) + \cos(x)$ at $x = \pi/4$.
4. Evaluate the indefinite integral: $\int (4x^3 - 1/x + e^x) dx$.
5. Find the area under the curve $y = x^2$ from $x = 0$ to $x = 2$.

